

# Good Data, Bad Data

Project: Helpful Classroom Sorter

## 1. Your mission

### Your learning target

I can check records against a stated format, unit and meaning before using them.

I can distinguish an evidence-based correction from an unsupported guess.

I can keep original values and a change log, then report what remains unresolved.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Coverage, correctness, and relevance matter more than raw quantity.



LEARNING TARGET A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

Learner name:

### Readiness for the connected explanation

A recommender needs availability, but that field is blank. Can the highest preference score establish that the item is available? Explain.

### This week's end goal

By the end, I can repair data with traceable evidence while preserving valid zero and unknown.

## 2. Notice the evidence

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



**MYSTERY** One animal dataset has 1,000 blurry duplicate photos; another has 120 clear, correctly labeled photos across settings and animal types.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

**GOOD THINKING** A prediction can change when stronger evidence appears.

### A closer explanation

A tidy table can still contain wrong information. Validation checks whether a record follows a schema. A schema says which fields are required and what their values mean. Passing these checks makes a record usable for a stated task; it does not prove that the measurement or claim is true.

Our fictional RampLog records one ramp trial per trial ID. The fields are trial ID, height in centimetres, travel in centimetres and source card. Height must be a whole number from 1 through 10. Travel must be a whole number from 0 through 10. The source card identifies the original observation. Repeated trial IDs need investigation before counting trials.

The purpose is to count trials whose travel is at least 4 cm. A measured zero means no travel; a blank means no value was recorded. Never silently replace a blank by zero. Use only the invented records here;

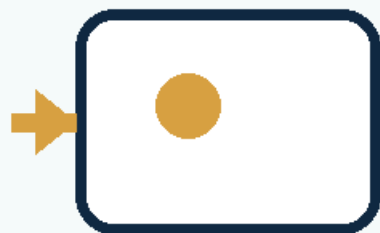
no personal information is needed.

### 3. See how it works

VISUAL MECHANISM

**Input** → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



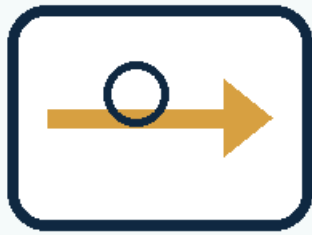
**INPUT**

collected examples and labels



**PROCESS**

audit coverage, correctness, relevance,  
and duplication



## OUTPUT

accept, repair, or reject the dataset

INPUT collected examples and labels

PROCESS audit coverage, correctness, relevance, and duplication

OUTPUT accept, repair, or reject the dataset

CORE IDEA Coverage, correctness, and relevance matter more than raw quantity.

UNDER THE HOOD Data quality is multidimensional. More rows can amplify duplicates, systematic labeling errors, stale conditions, or irrelevant fields. Fitness depends on the intended population and decision, not a universal quality score.

Draw a fresh example with all three stages:

### A closer explanation

#### Preserve the original and explain each decision

**Original record · Evidence check · Working decision**

**Original record:** T2 height 60 cm

**Evidence check:** Card C2 says 6 cm

**Working decision:** Correct to 6; log C2

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**Original record:** T3 travel blank

**Evidence check:** Card C3 also blank

**Working decision:** Keep missing; flag

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**Original record:** T4 travel 0 cm

**Evidence check:** Card C4 confirms zero

**Working decision:** Keep measured zero

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C2 supports correcting T2. C3 cannot supply a missing measurement. C4 shows that zero is a recorded observation, not a placeholder for missing data.

The complete raw table is T1: height 4, travel 3, card C1; T2: height 60, travel 5, card C2; T3: height 5, travel blank, card C3; T4: height 6, travel 0, card C4. A second export row repeats T1 with height 4, travel 3 and C1. C1 confirms one trial T1, not two. C2 reads height 6 and travel 5. C3 has no travel measurement. C4 reads height 6 and travel 0.

Work in a separate corrected copy. Record T2 height 60 to 6, evidence C2, transcription error, reviewer and review date. Link the repeated T1 export row to the first T1; count that single trial once and keep both raw rows. Do not remove two different trials merely because their measured numbers match.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

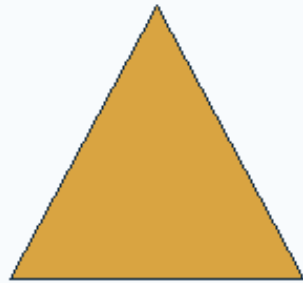
### PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



**1,000 duplicates**



**120 varied photos**



**Wrong labels**



**Owner birthday**

CASE 1 — 1,000 duplicates Clue: high count, low variety. Meaning: weak coverage.

CASE 2 — 120 varied photos Clue: settings and types represented. Meaning: stronger coverage.

CASE 3 — Wrong labels Clue: targets conflict with reality. Meaning: correctness failure.

CASE 4 — Owner birthday Clue: unrelated to animal type. Meaning: irrelevant/private.

BOUNDARY Large quantity cannot compensate for systematic absence or incorrect labels.

Which two pictures are easiest to confuse? What evidence separates them?



## Change one thing in the pictured case

Change: The source log gives B only as bare 200 with no unit.

Prediction: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





**correctness**



**relevance**



### Word · Meaning · My example

**Word:** coverage

**Meaning:** how well examples represent needed situations

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** correctness

**Meaning:** whether records and labels are accurate

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** relevance

**Meaning:** whether data supports the task

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** duplicate

**Meaning:** repeated record adding little new coverage

**My example:** \_\_\_\_\_

\_\_\_\_\_

### Misconception detective

CLAIM TO REPAIR The bigger dataset must be better.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

## Words used in the closer explanation

### Word · Meaning

**Word:** Schema

**Meaning:** A description of required fields, formats, units and allowed values

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**Word:** Validation

**Meaning:** Checking a record against stated requirements

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**Word:** Provenance

**Meaning:** A record of where information came from

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**Word:** Correction

**Meaning:** A change supported by evidence about the intended value

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**Word:** Change log

**Meaning:** A record of what changed, why, when and on what evidence

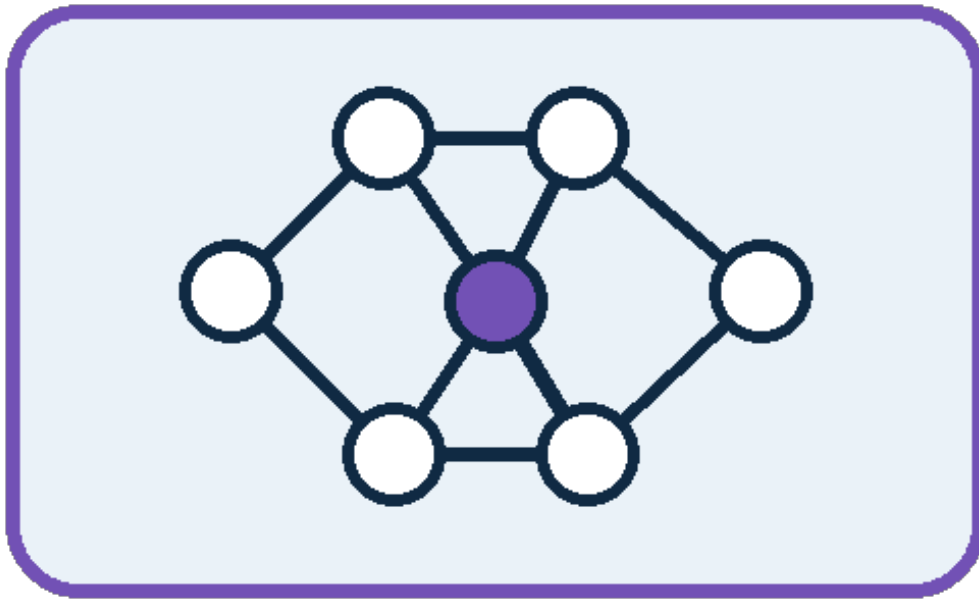
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## 6. Follow a worked example

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE One animal dataset has 1,000 blurry duplicate photos; another has 120 clear, correctly labeled photos across settings and animal types.

STEP 1 — OBSERVE collected examples and labels

STEP 2 — TRACE audit coverage, correctness, relevance, and duplication

STEP 3 — CONCLUDE accept, repair, or reject the dataset

MODEL REASONING Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

DO NOT OVERCLAIM Large quantity cannot compensate for systematic absence or incorrect labels.

What new evidence could change this answer?

### A closer explanation

**Problem · Permitted action here · Unsupported shortcut**

**Problem:** Invalid height 60

**Permitted action here:** Check C2; correct to 6

**Unsupported shortcut:** Guess by deleting a zero

**Problem:** Missing travel

**Permitted action here:** Keep missing; seek source

**Unsupported shortcut:** Fill with the class average

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**Problem:** Repeated T1 ID

**Permitted action here:** Confirm one trial on C1; count once

**Unsupported shortcut:** Treat every repeated number as a duplicate

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**Problem:** Two sources disagree

**Permitted action here:** Flag conflict; ask for resolution

**Unsupported shortcut:** Pick the value that improves the result

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A conversion changes units while preserving the quantity. If an authorised source records 50 mm, that is 5 cm because 10 mm equals 1 cm. Write the original unit, conversion and resulting unit in the log. A bare value 50 without a known unit does not justify assuming millimetres.

A plausible value can still be a transcription error: 7 would pass the height rule even if its source said 6. Conversely, an unusual but supported value is not automatically wrong. Checking format and checking agreement with a source are different jobs. A source can itself contain an error, so retain its identity and any unresolved concern.

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Good Data, Bad Data

A fictional toy-car distance table needs cleaning before use.

## The facts we will use

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

## Good Data, Bad Data

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Validate rows

A's zero is valid; B has wrong unit; C is missing; A is duplicated.



#### 2. Evidence repair

200 mm / 10 = 20 cm; source log supports retaining one A.



#### 3. Keep unknown

C stays unresolved; do not replace it with 0.



#### 4. Clean record

Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

### CHANGE ONE THING

The source log gives B only as bare 200 with no unit.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Validate rows

**What happens and why:** A's zero is valid; B has wrong unit; C is missing; A is duplicated.

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**Step:** 2. Evidence repair

**What happens and why:** 200 mm ÷ 10 = 20 cm; source log supports retaining one A.

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**Step: 3. Keep unknown**

**What happens and why:** C stays unresolved; do not replace it with 0.

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**Step: 4. Clean record**

**What happens and why:** Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a duplicate copy. Zero means a measured zero distance; blank means no supplied measurement. Keep the raw table and explain every change so someone can reconstruct what happened.

## What this example does not establish

A clean table can still contain measurement or labelling errors that these checks do not detect.

Read the labelled picture: Which repairs are evidence-backed in the original case, and which row must remain unknown?

Change one thing: The source log gives B only as bare 200 with no unit. Predict the result before checking the explanation. What remains the same?

# 7. Find and repair a mistake

COMPARE

## Same Result, Different Reason

A comparison helps reveal the true concept boundary.

### Case · Evidence · What it shows

**Case:** 1,000 duplicates

**Evidence:** high count, low variety

**What it shows:** weak coverage

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**Case:** 120 varied photos

**Evidence:** settings and types represented



**What it shows:** stronger coverage

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**Case:** Wrong labels

**Evidence:** targets conflict with reality

**What it shows:** correctness failure

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**Case:** Owner birthday

**Evidence:** unrelated to animal type

**What it shows:** irrelevant/private

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## Repair two tempting claims

CLAIM A The bigger dataset must be better.

Repair:

CLAIM B Clean formatting means the data is correct.

Repair:

STRONG ANSWER FRAME I claim \_\_\_\_\_ because the evidence shows \_\_\_\_\_. The process is \_\_\_\_\_. One limit is \_\_\_\_\_.

## A closer explanation

After changing a value, run the schema checks again and compare the corrected copy with the raw copy and log. Recount distinct trials. Check that every correction has evidence and that zero, missing and conflict remain separate. A repair can introduce a new mistake if a value is changed in the wrong row.

The corrected RampLog supports a statement about its three assessed trials. It does not establish how every ramp or future trial behaves. Better formatting cannot create missing observations or broaden what was sampled. Next week asks which cases a dataset actually covers.

**A table now has no blank cells because a designer guessed all missing measurements. Explain why this does not establish better evidence, and state what a trustworthy repair record should retain**

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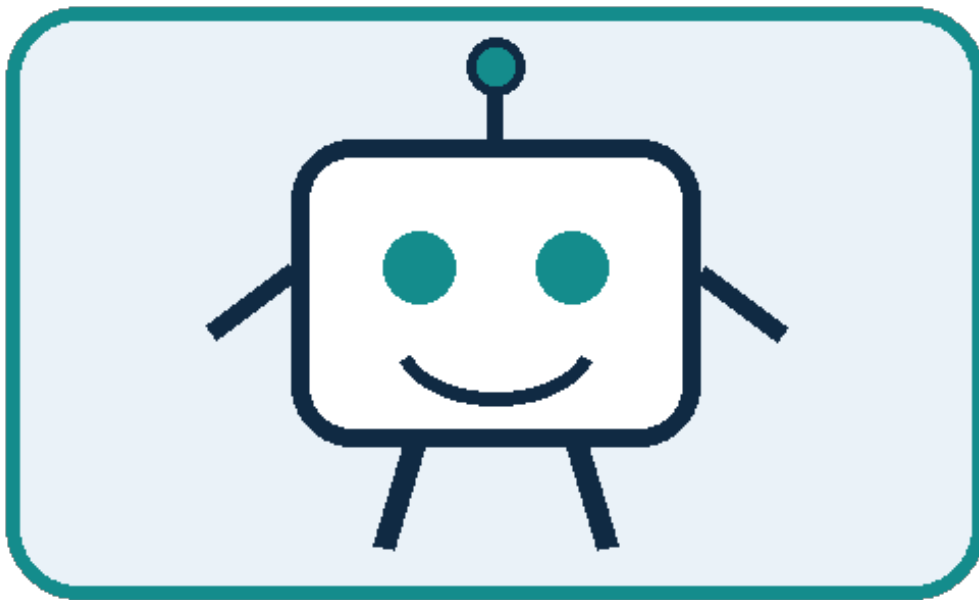
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## 8. Practise together

### GUIDED PRACTICE

#### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Coverage, correctness, and relevance matter more than raw quantity.

Use a fresh example to explain coverage.

Trace a new input → process → output example.

Find and repair the error in: Clean formatting means the data is correct.

Invent one edge case and name the safe response.

**PARTNER CHECK** Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

After the documented correction and confirmed duplicate handling, RampLog contains four distinct trials: T1, T2, T3 and T4. Three have recorded travel. T1 has 3 cm, T2 has 5 cm and T4 has 0 cm. Only T2 meets at least 4 cm, so the result is 1 of 3 assessed trials, with 1 trial missing travel. Do not report T3 as a

failure or hide it from the account.

**A reviewer fills T3 travel with 4 because it makes the results look more balanced. Identify the unsupported change, write an honest report using the recorded evidence, and name the source information still needed**

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Describe which records entered the calculation and why. Keep unresolved records visible in a separate status list. A new measurement is a new observation with its own time and identity; it should not be passed off as the missing observation from an earlier trial.

### **Practise with less help: Write a traceable correction**

Use workbook W2 for the connected example “Data quality and documented repair”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### **Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### **Check your reasoning**

Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### **Explain the picture**

Which repairs are evidence-backed in the original case, and which row must remain unknown?

**My explanation using a pictured step and the given facts**

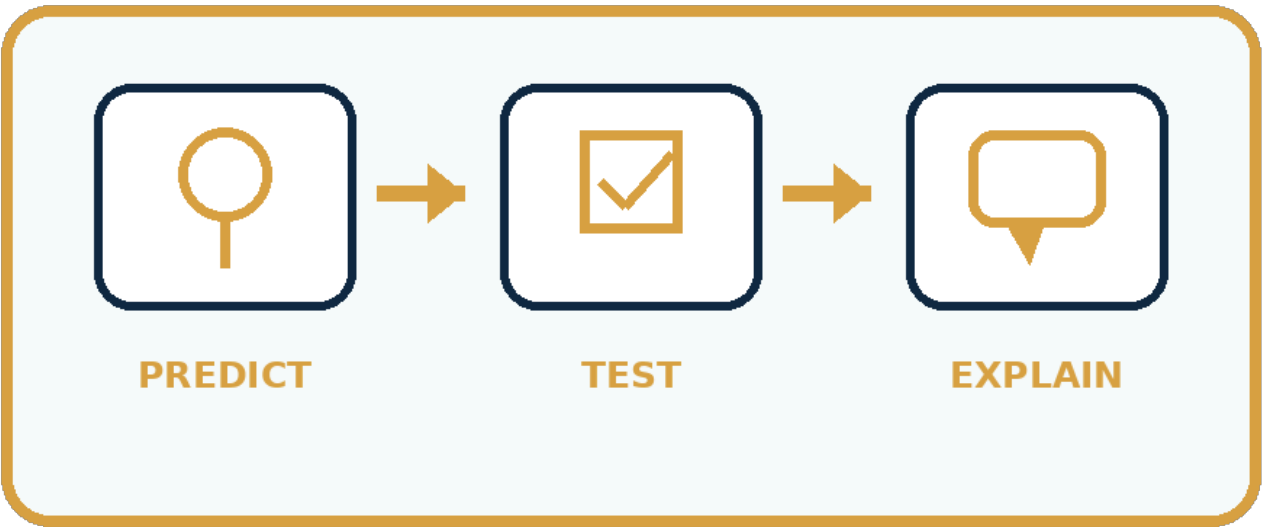
This is supported practice. Keep the separately named independent case for an attempt before feedback.

# 9. Run the investigation

EXPLORER LAB

## Dataset Quality Audit

Predict first, test honestly, and explain what the evidence means.



- State your prediction before the result is known.
- Complete the task: Audit two mini datasets and choose the safer one.
- Keep a normal case and an edge case clearly separated.
- Record what happened—even when it does not match your prediction.
- Explain the result, one limitation, and the next useful test.

**Case · Prediction · Observed result · What it means**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Edge

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

\_\_\_\_\_

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

**Plan → try → check**

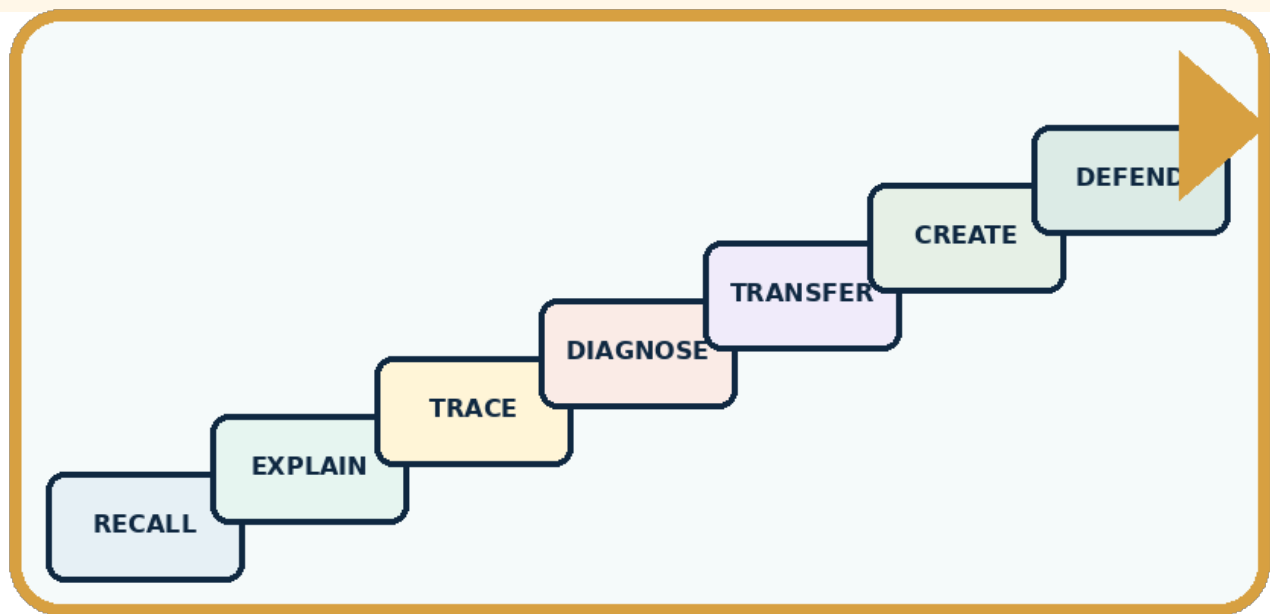
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define coverage.

EXPLAIN Coverage, correctness, and relevance matter more than raw quantity.

TRACE Show the input, process, and output.

DIAGNOSE Repair: The bigger dataset must be better.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

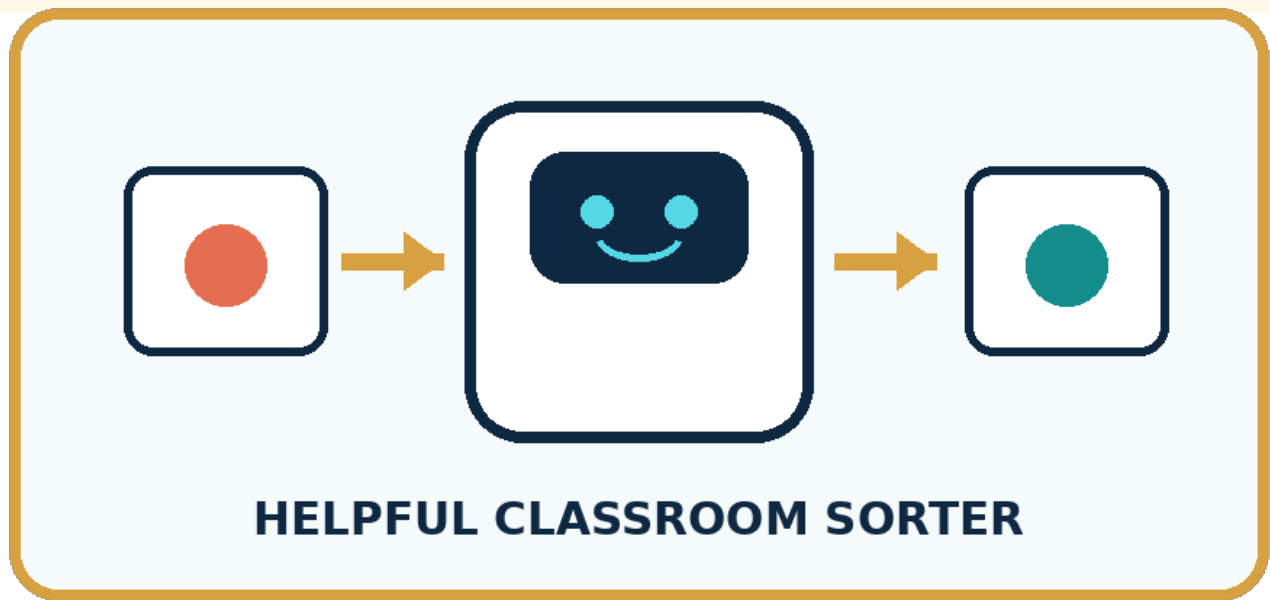
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

## 11. Connect your project

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



**THIS WEEK'S CHECKPOINT** Audit the sorter's example cards and add a reason for including each field and condition.

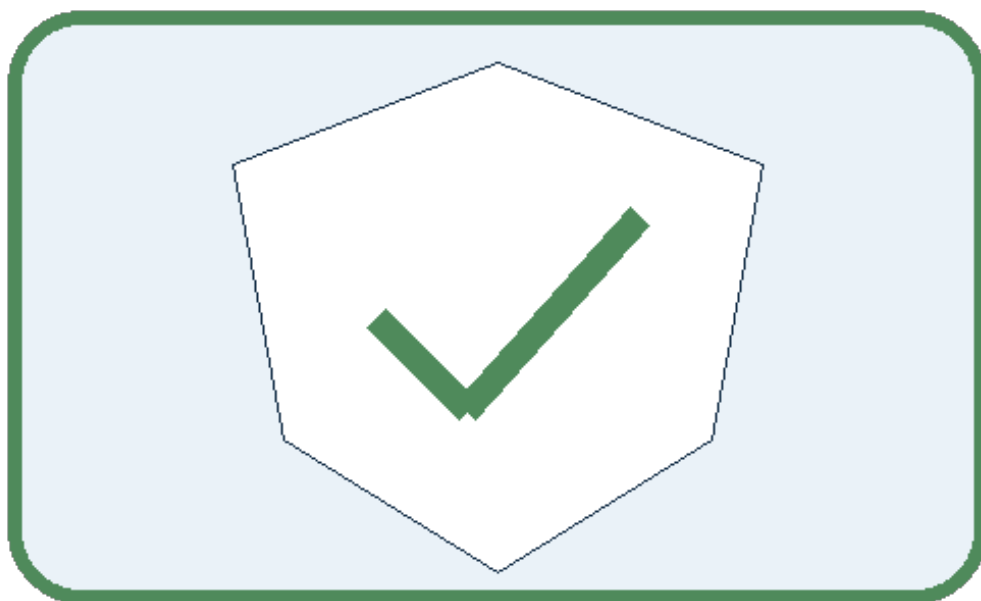
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Large quantity cannot compensate for systematic absence or incorrect labels.

My project decision:

### Evidence for your project

Keep raw sorter inputs, validated records and an evidence-linked repair log as different artifacts.

## 12. Show what you know

MASTERY

### Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



REQUIRED PROOF Audit two mini datasets using coverage, correctness, relevance, and duplication; choose the safer one with evidence.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.



Independent evidence:

**SELF-CHECK** Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

**NEXT BRIDGE** Carry today's evidence habit into: Missing and Unbalanced Examples.

## 13. Study a complete case

### Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

#### The question

What needs checking? May 130 silently become 13?

#### Worked explanation

Investigate missing and unusually large values using the source and units. Do not invent a repair. Document verified corrections.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### Check suspicious data before changing it

**Evidence or stage · Value or result**

**Evidence or stage:** 12 cm

**Value or result:** Plausible recorded length

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**Evidence or stage:** 14 cm

**Value or result:** Plausible recorded length

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**Evidence or stage:** Blank

**Value or result:** Missing measurement

---

**Evidence or stage:** 130 cm

**Value or result:** Check source and units

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Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

What needs checking? May 130 silently become 13?

Investigate missing and unusually large values using the source and units. Do not invent a repair.  
Document verified corrections.

## 14. Try a new case independently

### Independent case: Audit an independent paper-plane log

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank. S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Audit the sorter's example cards and add a reason for including each field and condition.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional toy-car distance table needs cleaning before use.

**By the end, I can · How I will show it**

**By the end, I can:** Check records against a stated format, unit and meaning before using them

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Distinguish an evidence-based correction from an unsupported guess

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Keep original values and a change log, then report what remains unresolved

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

# 16. Finish and return

Close your notes. Explain this goal in your own words: Check records against a stated format, unit and meaning before using them

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.